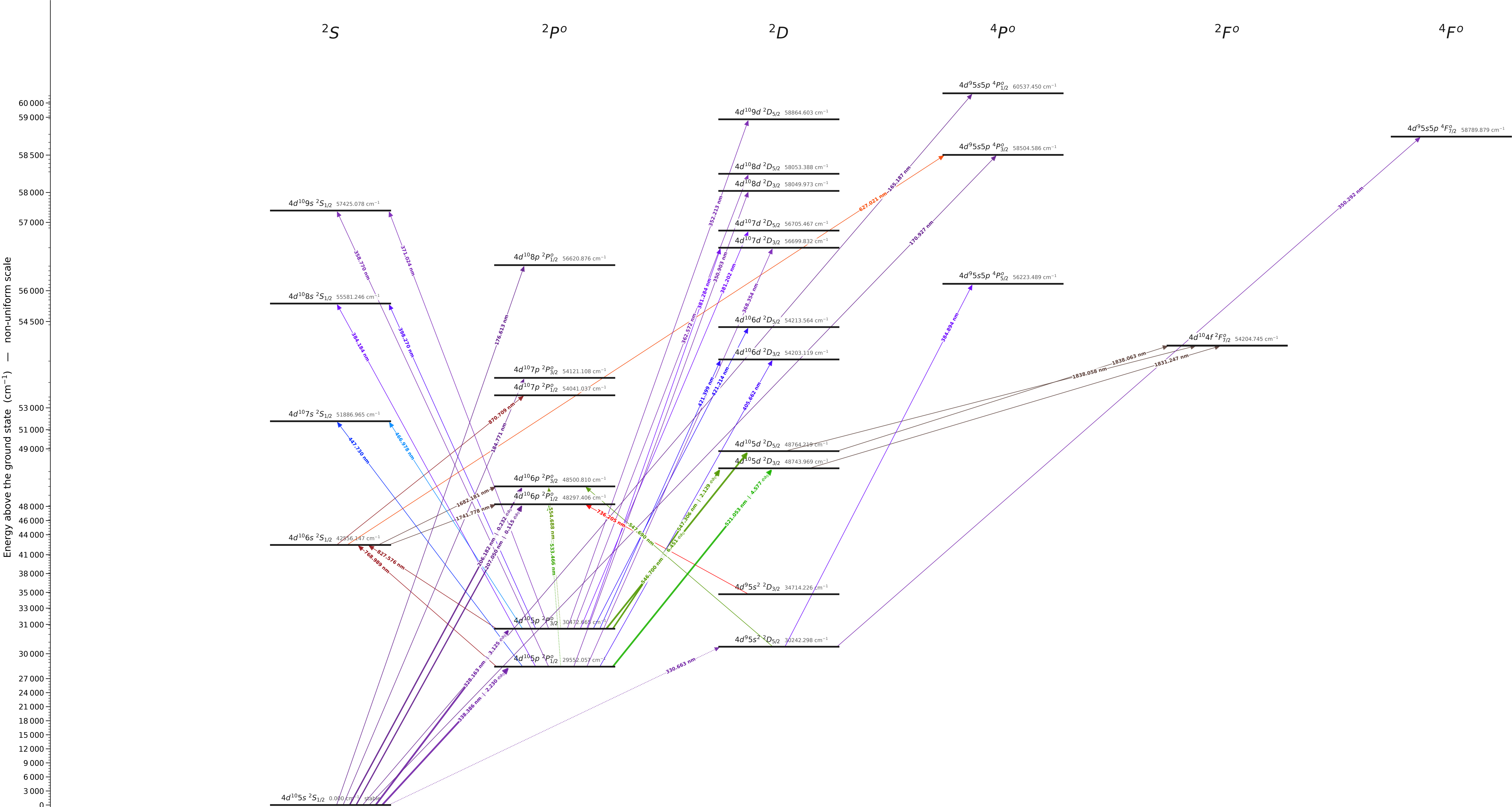
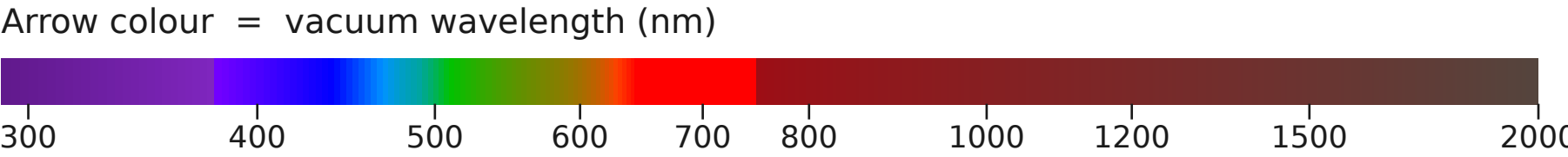


Silver

Ag (Ag I). Strongest classified transitions below 2 μm (NIST): 29 levels, 42 lines.

Ag ⁺ ionisation limit 61106.45 cm⁻¹ = 7.57623 eV (163.649 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $| \langle J || e r || J' \rangle |$ in ea₀
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, n = 70.
Level caption: NIST energy and measured lifetime (where one exists).



Strongest transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	⟨J er J'⟩ (ea ₀)	A (s ⁻¹)	Γ/2π up. (MHz)	branch (%)	Use
4d ¹⁰ 5s ² S _{1/2} - 4d ¹⁰ 6p ² P _{3/2} ^o	206.1821	206.1162	1454.0177	0.232	3.100e+06	-	-	-
4d ¹⁰ 5s ² S _{1/2} - 4d ¹⁰ 6p ² P _{1/2} ^o	207.0505	206.9844	1447.9198	0.115	1.500e+06	-	-	-
4d ¹⁰ 5s ² S _{1/2} - 4d ¹⁰ 5p ² P _{3/2} ^o	328.1630	328.0684	913.5475	3.125	1.400e+08	-	-	-
4d ¹⁰ 5s ² S _{1/2} - 4d ¹⁰ 5p ² P _{1/2} ^o	338.3859	338.2888	885.9484	2.230	1.300e+08	-	-	-
4d ¹⁰ 5p ² P _{1/2} ^o - 4d ¹⁰ 5d ² D _{3/2}	521.0528	520.9078	575.3590	4.577	7.500e+07	-	-	-
4d ¹⁰ 5p ² P _{3/2} ^o - 4d ¹⁰ 5d ² D _{5/2}	546.7004	546.5485	548.3670	6.451	8.600e+07	-	-	-
4d ¹⁰ 5p ² P _{3/2} ^o - 4d ¹⁰ 5d ² D _{3/2}	547.3063	547.1543	547.7599	2.129	1.400e+07	-	-	-

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
Γ/2π: natural linewidth of the upper level, 1/(2πτ).